January 2026 CSE 216
Online 1: Basic SQL (B1+B2)

Time: 1h + Submission: 5 min

<table>
<tr><th>Sl.</th><th>Questions</th></tr>
<tr><td>1.</td><td>Select Employee_ID, Full Name, Job_ID and Salary of employees whose first name starts with 'M', ends with ‘k’ and has at least 4 characters.</td></tr>
<tr><td>2.</td><td>From the EMPLOYEES table, find all employees who have the letter 'e' (ignoring uppercase or lowercase) in their first name, where the position of the very first 'e' counting forward from the start of their first name perfectly matches the position of an 'e' when counting backward from the end of their last name. Additionally, filter the results so you only include employees who were hired in the second half of any year (from July through December). For your final output, display the employee's full name (first and last name separated by a space) and their email address. To solve this, you cannot use the less-than (<) and greater-than (>) operators.</td></tr>
<tr><td>3.</td><td>Write a single query to display the department_id and the total salary expenses (aliased as total_qualified_payroll) for each department. However, your query must only include employees whose 10-year work anniversary falls in the first half of a calendar year (between January 1st and June 30th).</td></tr>
<tr><td>4.</td><td>i. Create a table named projects with the following structure:
<table>
<tr><th>Column</th><th>Data type</th></tr>
<tr><td>project_id</td><td>NUMBER(4)</td></tr>
<tr><td>project_name</td><td>VARCHAR2(50)</td></tr>
<tr><td>budget</td><td>Number(10, 2)</td></tr>
<tr><td>start_date</td><td>DATE</td></tr>
<tr><td>end_date</td><td>DATE</td></tr>
<tr><td>status</td><td>CHAR(1)</td></tr>
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	ii. Add the following constraints: ● project_id is the primary key. ● project_name must be unique. ● budget must be greater than 50000. ● status can contain only 'N' (New), 'A' (Active), or 'F' (Finished). iii. Add a new column named manager_name VARCHAR2(40)
5.	i. Add a new column to Employees table named is_hired_in_first_half. This column can have only two values 'Y', 'N'. ii. Set is_hired_in_first_half = 'Y' if the employee is hired in first half of the year (from 01 Jan to 30 June) and 'N' otherwise. iii. Add a new column to the Employee table named work_email. For each employee, set their work_email to be first_name.department_acronym.hire_year@company.com (all lowercase). department_acronym is the part in job_id before the underscore.